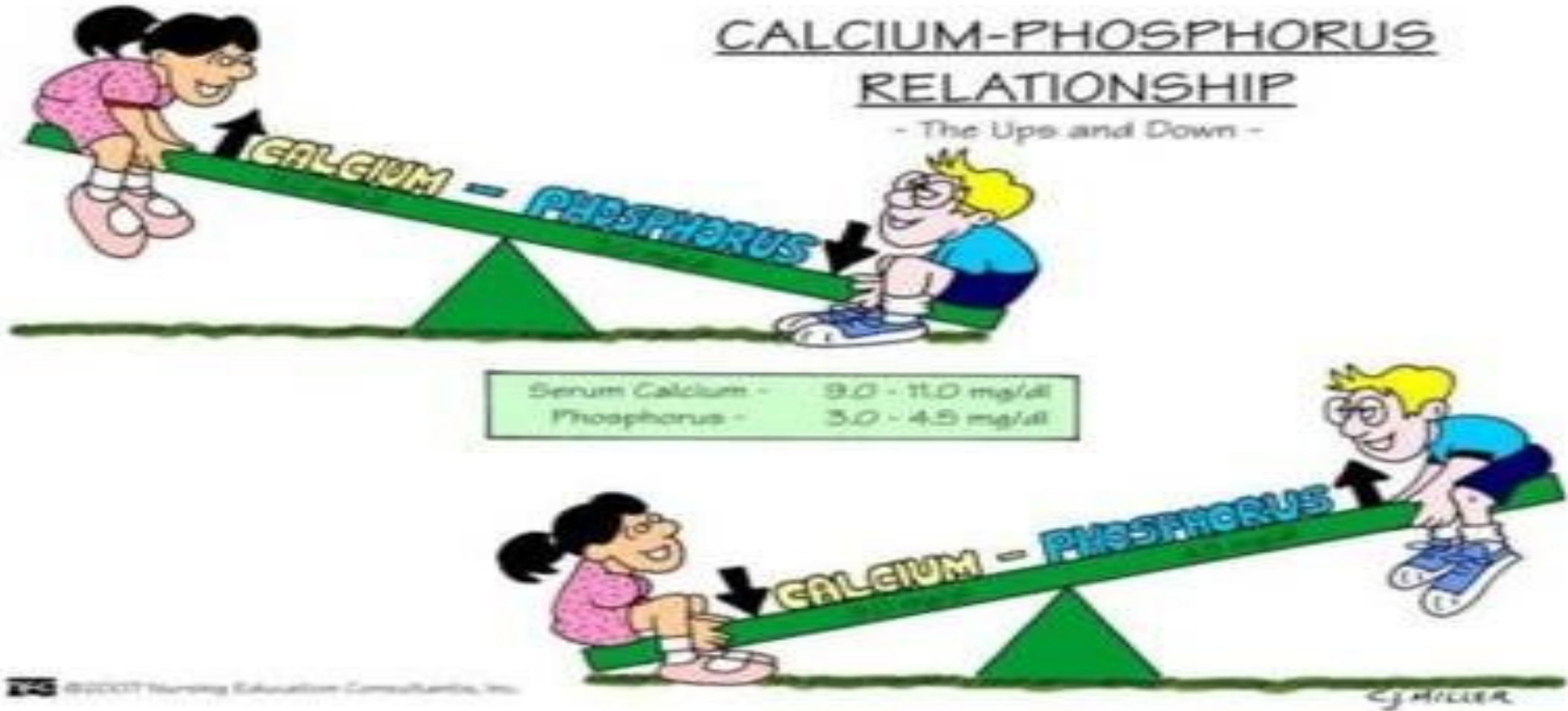


Calcium and phosphate balance



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Calcium and phosphate balance

- **Explain** regulation of calcium and phosphate
- **Describe** hormonal control mechanisms
- **Relate** balance to physiological functions
- Functions of Calcium and Phosphorus
- Dietary sources of Calcium and Phosphate
- Factors regulating serum calcium and phosphate levels

Calcium

- ❑ Calcium is the most abundant element in the body.
- ❑ Calcium is the fifth most required element by the human body after oxygen, carbon, hydrogen, and nitrogen.
- ❑ It contributes 1.9% to body weight.
- ❑ About 98% of the calcium in the human body is found in bones.
- ❑ The remaining 1% is in teeth and other soft tissues and only 0.1% is in the extracellular fluid (ECF) Which is involved in a number of processes such as signaling ,muscle contraction and blood clotting.
- ❑ Calcium is reabsorbed from the 1st and 2nd part of the duodenum

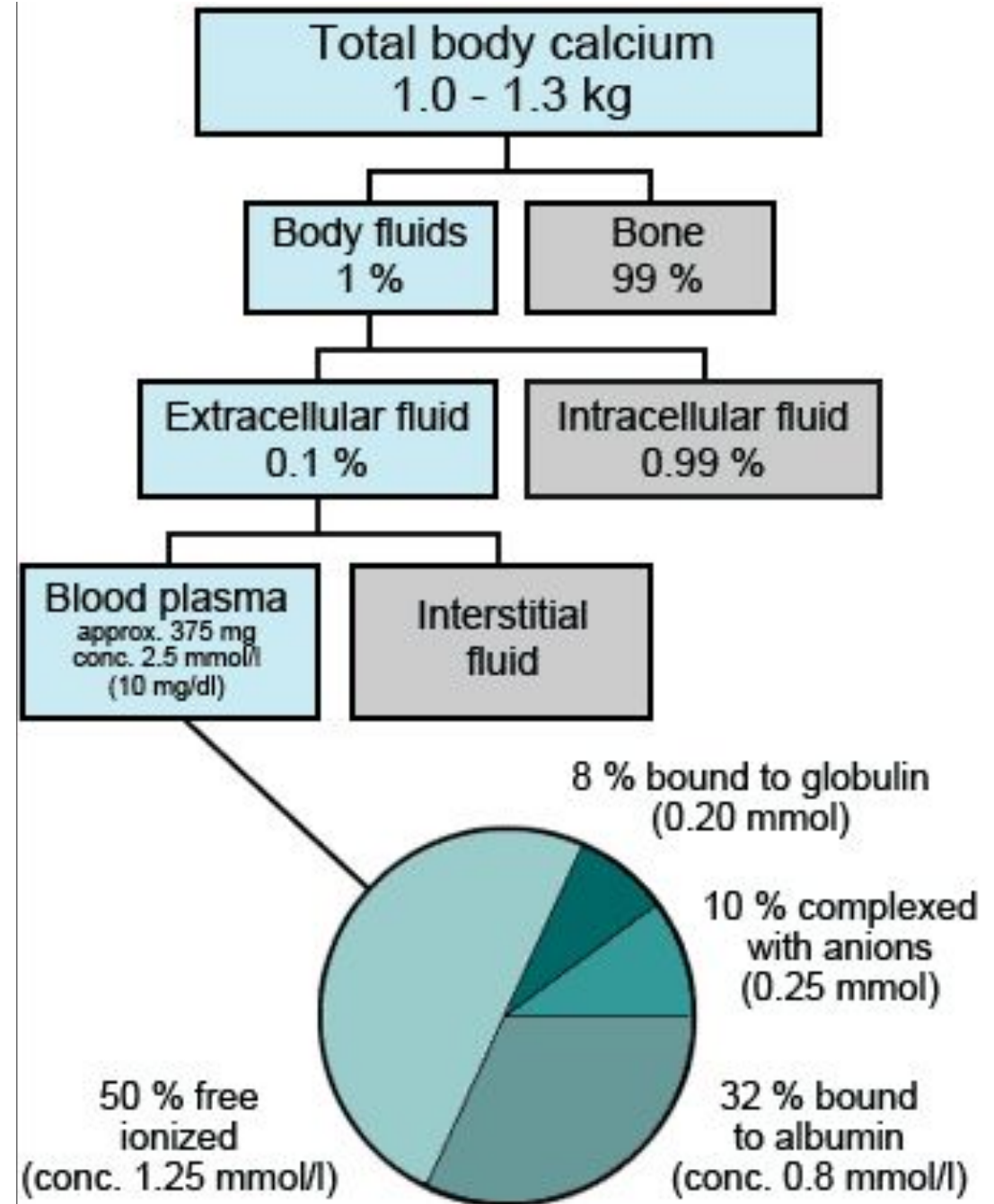
NORMAL VALUE

In a normal young healthy adult, there is about 1,100 g of calcium in the body.

It forms about 1.5% of total body weight. 99% of calcium is present in the bones and teeth and the rest is present in the plasma.

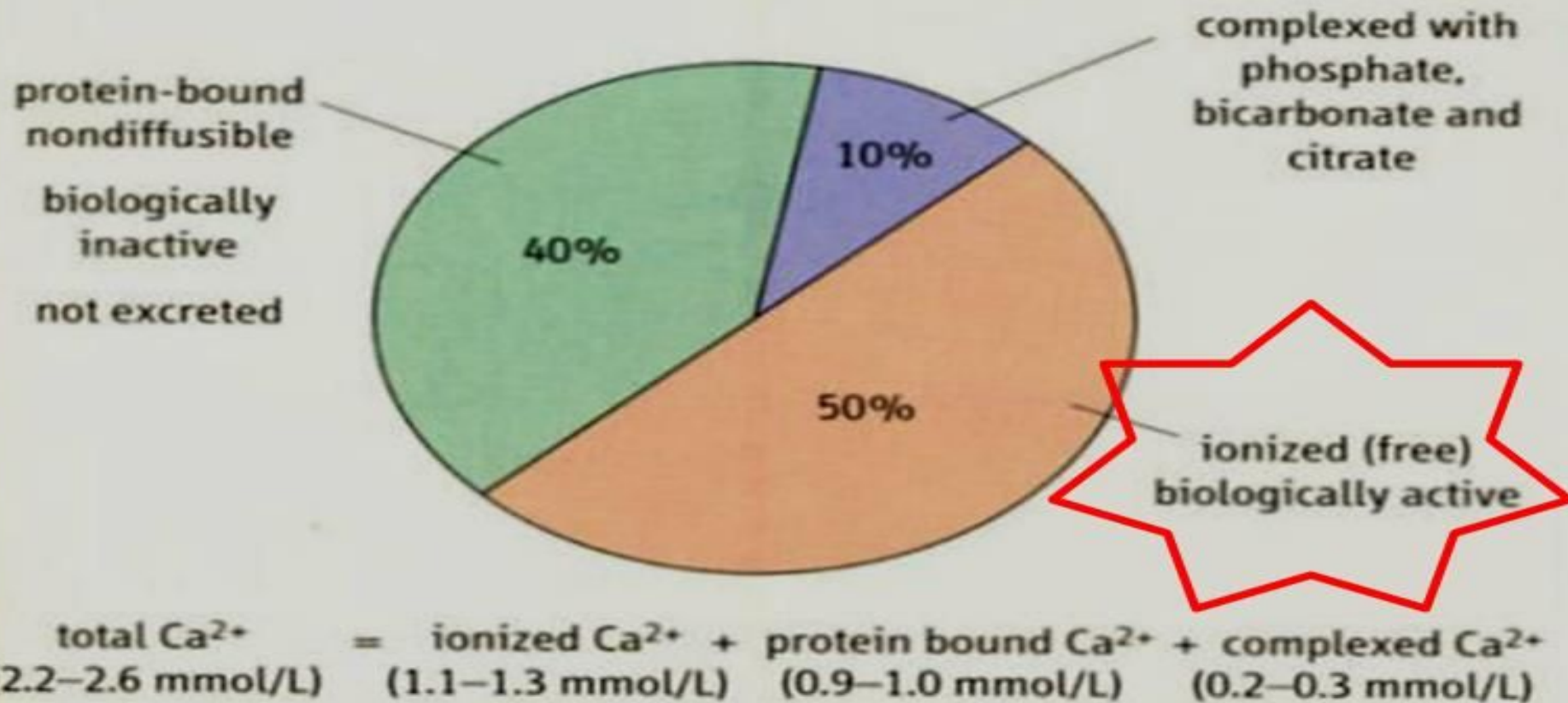
Normal blood calcium level ranges between 9 and 11 mg/d L.

(Mg/d L means milligrams per deciliter, which is a standard unit of measurement used in medical tests to show the concentration of a substance in a specific amount of fluid.)



Calcium in Blood

Circulating calcium fractions



SOURCE OF CALCIUM

1. Dietary Source

Calcium is available in several foodstuffs. Percentage of calcium in different food substance is:

- Whole milk = 10%
- Low fat milk = 18%
- Cheese = 27%
- Other dairy products = 17%
- Vegetables = 7%
- Other substances such as meat, egg, grains, sugar, coffee, tea, chocolate, etc. = 21%

2. From Bones

- Besides dietary calcium, blood also gets calcium from bone by resorption.

CALCIUM RICH FOODS

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FIRM TOFU



SKIM MILK



LOW-FAT
YOGURT



GRATED
PARMESAN
CHEESE



COOKED
SPINACH



MEDINAZ.COM



GREEN LEAFY
VEGETABLES



SHRIMP



ALMONDS



DRIED FIGS



FORTIFIED
CEREALS

Calcium Cheat Sheet



Spinach

30 mg

Serving: 1 Cup



Milk

250 mg

Serving: 200ml



Apple

13.4 mg

Serving: 1 Large



Broccoli

47 mg

Serving: 1 Cup



Dried figs

162 mg

Serving: 100 g



Oranges

74 mg

Serving: 1 Large



Almonds

80 mg

Serving: 30 g



Sesame

88 mg

Serving: 1tbsp



Dahi

85 mg

Serving: 100 g



Banana

6 mg

Serving: 1 Large



Paneer

480 mg

Serving: 100 g



Chia seeds

76 mg

Serving: 1tbsp

Foods High in *Calcium*



1 cup milk, calcium-fortified soy or almond milk



1 cup yogurt



1 1/2 ounces low-fat cheese



1 cup cottage cheese



3/4 cup cooked quinoa



1 medium fig



1/2 cup tofu



1 tablespoon balackstrap molasses



3oz Sardines (with bones)



3oz canned salmon (with bones)



1 cup calcium-fortified cereal



1/2 cup cooked amaranth



1 cup edamame



1 cup cooked soybeans



1/4 cup almonds



1/2 cup cooked greens



1/2 cup cooked broccoli



1 medium sweet potato



1 medium orange



1 slice calcium-fortified bread

Daily Calcium Requirements

Different stages. Different needs.

Stronger bones
at every age



Infants 0 – 12 months	Children 1 – 8 years	Teens 9 – 18 years	Adults 19 – 50 years	Women 51 – 70 years	Men 51 – 70 years	Adults 70+ years	Pregnant / Nursing (19 – 50 years)
							
200 – 260 mg	700 – 1,000 mg	1,300 mg	1,000 mg	1,200 mg	1,000 mg	1,200 mg	1,000 – 1,300 mg



Builds strong bones
and teeth



Helps with muscle,
nerve and heart function



Meets higher needs
at key life stages

Calcium requirement according to age

Age Group	Calcium Requirements
1-3 years	500mg/day
4-8 years	700mg/day
9-11 years	1000mg/day
12-18 years	1300mg/day
19+ years	1000mg/day
Women 50+ years	1300mg/day
Men 70+ years	1300mg/day

Importance and Functions of Calcium

Calcium is very essential for many activities in the body such as:

1. Bone and teeth formation
2. Neuronal activity □ Calcium plays a pivotal role in nerve impulse transmission.
3. Skeletal muscle activity □ Calcium salts provide rigidity to the skeleton.
4. Cardiac activity
5. Smooth muscle activity
6. Secretory activity of the glands and secretion of some hormones is dependent on calcium.
7. Cell division and growth
8. It helps in Coagulation of blood

IMPORTANT FUNCTIONS OF CALCIUM IN OUR BODY

NERVE IMPULSE
TRANSMISSION

MAINTAIN BLOOD
PRESSURE

IMPORTANT ROLE IN
MUSCLE CONTRACTION

BLOOD CLOTTING

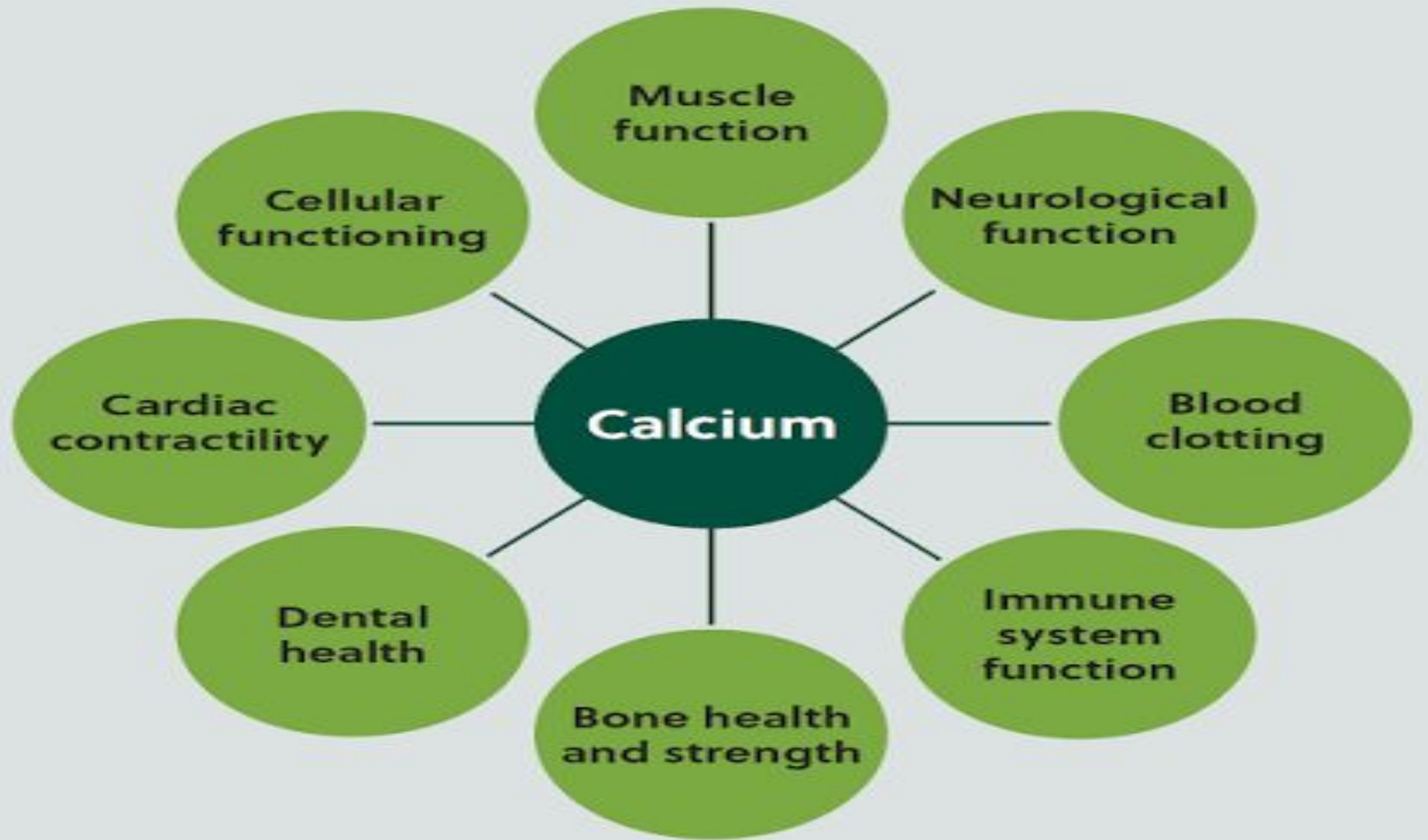
DENTAL HEALTH

BONE HEALTH

COMBAT KIDNEY
STONE AND DISEASE

REDUCES RISK OF
CANCER SUCH AS CRC,
BREAST CANCER





REGULATION OF BLOOD CALCIUM LEVEL

Blood calcium level is regulated mainly by three hormones

1. Parathormone

2. 1,25-dihydroxycholecalciferol (calcitriol)

3. Calcitonin.

1. Parathormone

Parathormone is a protein hormone secreted by parathyroid gland and its main function is to increase the blood calcium level by mobilizing calcium from bone (resorption)

2. 1,25-dihydroxycholecalciferol – Calcitriol

Calcitriol is a steroid hormone synthesized in kidney. It is the activated form of vitamin D. Its main action is to increase the blood calcium level by increasing the calcium absorption from the small intestine

3. Calcitonin

Calcitonin secreted by parafollicular cells of thyroid gland. Thyroid gland is a calcium-lowering hormone it reduces the blood calcium level mainly by decreasing bone resorption.

Effects of Other hormones

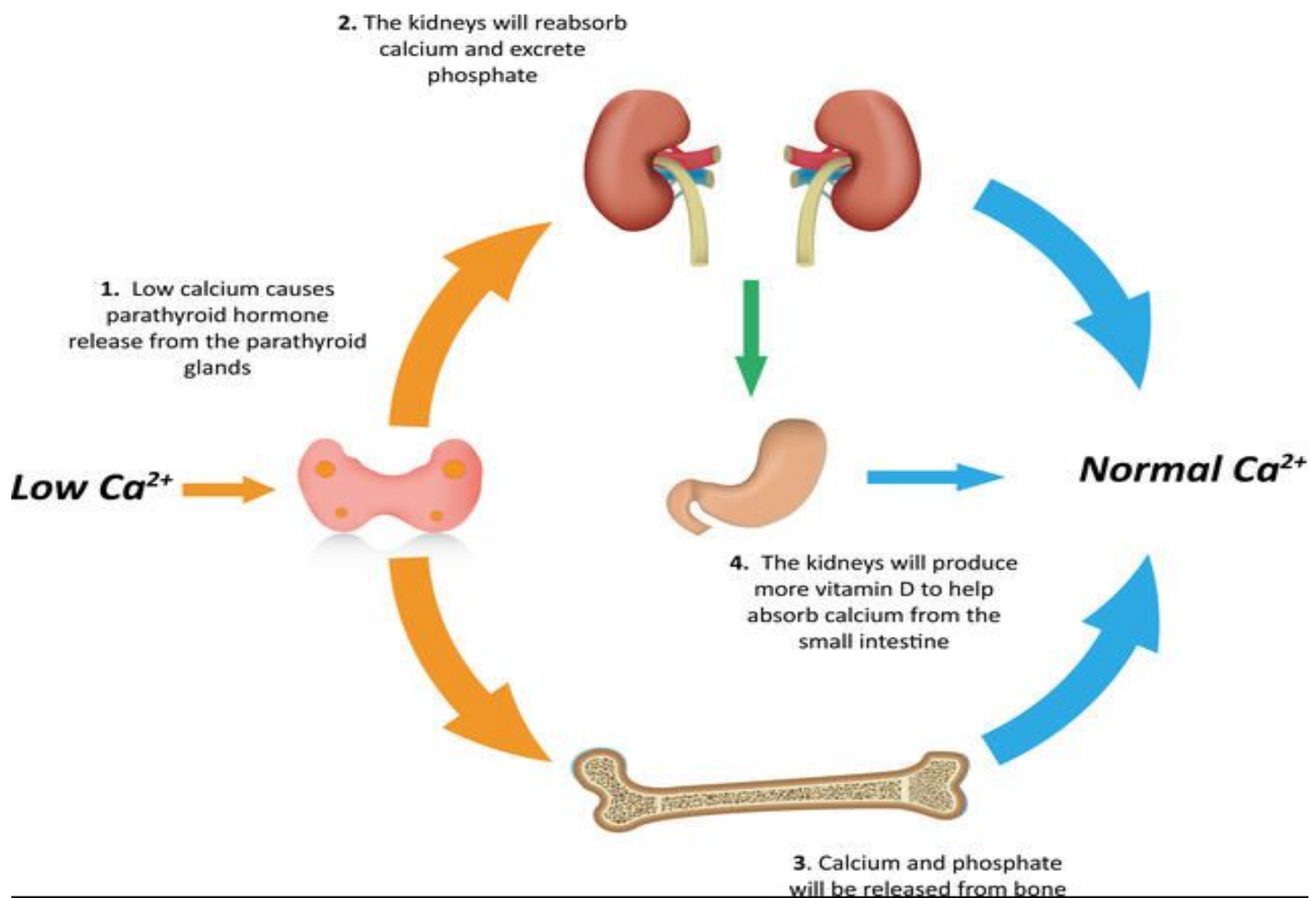
Growth hormone and Glucocorticoids also influence the calcium level

1. Growth hormone

Growth hormone increases the blood calcium level by increasing the intestinal calcium absorption. It is also suggested that it increases the urinary excretion of calcium. However, this action is only transient.

2. Glucocorticoids

Glucocorticoids (cortisol) decrease blood calcium by inhibiting intestinal absorption and increasing the renal excretion of calcium.



ABSORPTION AND EXCRETION OF CALCIUM

Calcium taken through dietary sources is absorbed from GI tract into blood and distributed to various parts of the body. Depending upon the blood level, the calcium is either deposited in the bone or removed from the bone (resorption).

Calcium is excreted from the body through urine and feces.

Absorption from Tract Gastrointestinal tract

Calcium is absorbed from duodenum by carrier mediated active transport and from the rest of the small intestine, by facilitated diffusion. Vitamin D is essential for the absorption of calcium from GI tract.

Excretion

While passing through the kidney, large quantity of calcium is filtered in the glomerulus.

From the filtrate 98% to 99% of calcium is reabsorbed from the renal tubules into blood. Only a small quantity is excreted through urine.

Most of the filtered calcium is reabsorbed in the **distal convoluted tubules and proximal part of collecting duct.**

In distal convoluted tubule, parathormone increases the reabsorption.

In collecting duct, vitamin D increases the reabsorption and calcitonin decreases reabsorption.

About 1,000 mg of calcium is excreted daily. Out of this, 900 mg is excreted through feces and 100 mg through urine.

Phosphate

Total body Phosphorus is 500-800 g.

Approximately 85 % of the body's Phosphate is stored in bones.

14 to 15 % is stored in the cells and less than 1 percent in the extracellular fluids.

NORMAL VALUE

Total amount of phosphate in the body is 500 to 800 g.

Though it is present in every cell of the body, 85% to 90% of body's phosphate is found in the bones and teeth.

Normal plasma level of phosphate is 4 mg/dL.

Sources of Phosphorus



Roe, mixed



Salmon



Mussels



Lentils



Soybeans



Greek yogurt



Various cheese types



Pork loin



Chicken meat



Beans

IMPORTANCE OR FUNCTIONS OF PHOSPHATE

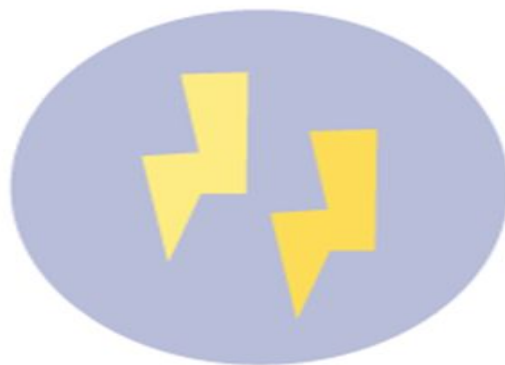
1. Phosphate is an important component of many organic substances such as, ATP, DNA, RNA and many intermediates of metabolic pathways
2. Along with calcium, it forms an important constituent of bone and teeth
3. It forms a buffer in the maintenance of acid-base balance.
4. It is an essential element for all living cells and important for energy utilization in the body.
5. It is present in the body in the form of phosphates (PO_4). It forms calcium phosphate with calcium and forms the skeleton of the bones.
6. It forms a buffer in the maintenance of acid-base balance.

functions of phosphorus

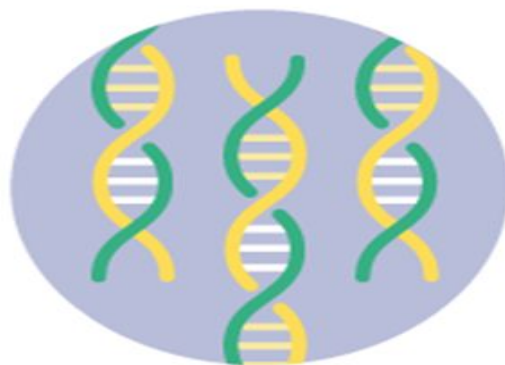
eufic.org



**forms and
keeps strong
bones and teeth**



**helps maintain
our energy levels**



**forms our genetic
material (DNA & RNA)**



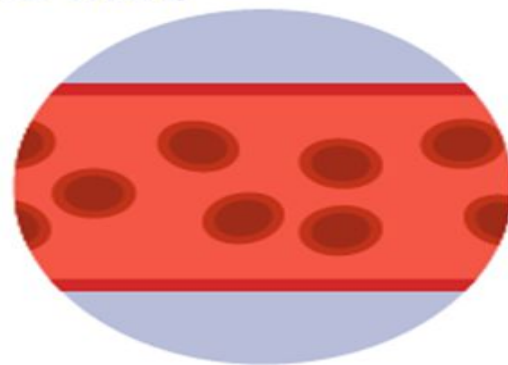
**helps form
the membranes
of our cells**



**keeps the healthy
function of our brain
& nervous system**



**helps to maintain
normal acid-
balance (pH)**



**helps oxygen
delivery in the body**

PHOSPHATE METABOLISM

Phosphorus (P) is an essential mineral that is required by every cell in the body for normal function.

Phosphorus is present in many food substances, such as peas, dried beans, nuts, milk, cheese and butter.

Inorganic phosphorus (P_i) is in the form of the phosphate (PO_4).

The majority of the phosphorus in the body is found as phosphate.

Phosphorus is also the body's source of phosphate.

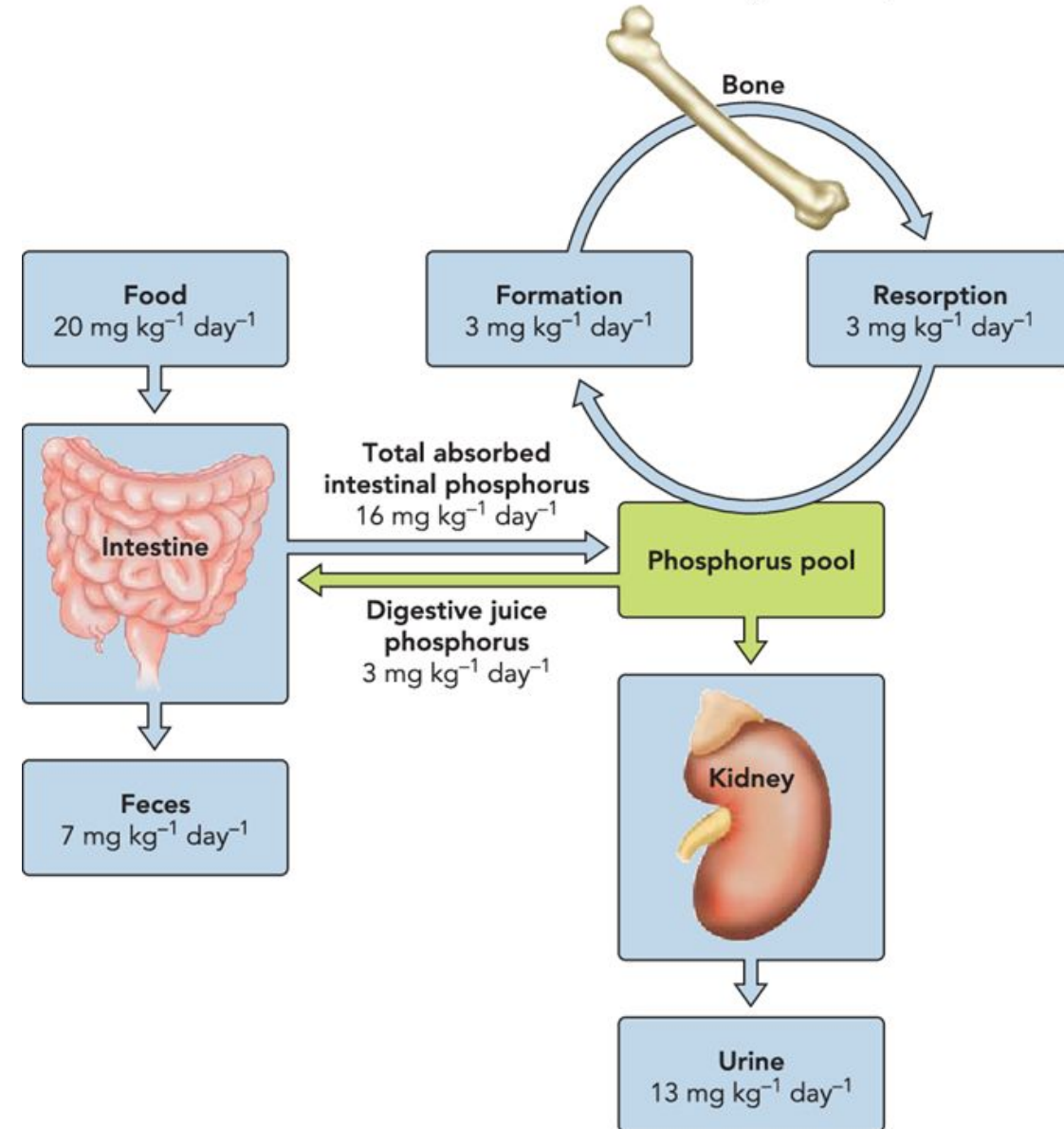
In body, phosphate is the most abundant intracellular anion.

REGULATION OF PHOSPHATE LEVEL

Phosphorus is taken through dietary sources. It is absorbed from GI tract into blood. It is also resorbed from bone. From blood it is distributed to various parts of the body. While passing through the kidney, large quantity of phosphate is excreted through urine.

Blood phosphate level is regulated mainly by three hormones:

1. Parathormone
2. Calcitonin
3. 1,25-dihydroxycholecalciferol (calcitriol)



1. Parathormone

It stimulates resorption of phosphate from bone and increases its urinary excretion. It also increases the absorption of phosphate from gastrointestinal tract through calcitriol. The overall action of parathormone is to decrease the plasma level of phosphate.

2. Calcitonin

Calcitonin also decreases the plasma level of phosphate by inhibiting bone resorption and stimulating the urinary excretion.

3. 1,25-Dihydroxycholecalciferol – Calcitriol

Calcitriol hormone increases absorption of phosphate from small intestine

Effects of Other Hormones

In addition to the previously mentioned three hormones, growth hormone and glucocorticoids also influence the phosphate level.

1. Growth hormone

Growth hormone increases the blood phosphate level by increasing the intestinal phosphate absorption.

2. Glucocorticoids

Glucocorticoids (cortisol) decrease blood phosphate by inhibiting intestinal absorption and increasing the renal excretion of phosphate.

Disorders of Calcium and Phosphorus

Hypocalcemia

- It is an abnormally low calcium level in the blood.

It can be caused by low levels of PTH,

- vitamin D deficiency,
- kidney failure,
- acute pancreatitis or
- insufficient magnesium and protein in the blood.

Table. Common Causes of Hypocalcemia

- Hypoparathyroidism
- Vitamin D deficiency or insufficiency
- Altered vitamin D metabolism due to medication usage
- Diseases affecting the kidneys and/or the liver
- Pseudohypoparathyroidism
- Hypomagnesemia or hypermagnesemia
- Hungry bone syndrome
- Infusion of phosphate
- Rapid citrated blood transfusion
- Medications

Calcium Deficiency Symptoms



BACKGROUND

- * SYMPTOM CHARACTERIZED by **INVOLUNTARY CONTRACTION** of MUSCLES
 - ~ CRAMPS, SPASMS, & SENSORY DISTURBANCES
- * CLASSIC SIGN of HYPOCALCEMIA
 - ↳ $\downarrow\downarrow$ RESTING MEMBRANE POTENTIAL of NEURONS
 - ↳ $\uparrow\uparrow$ NEUROMUSCULAR EXCITABILITY

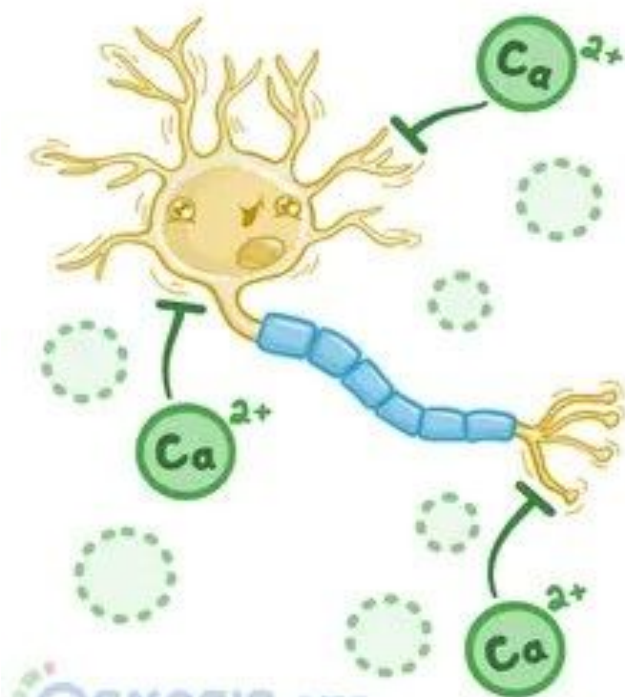


CAUSES

- * USUALLY a COMBINATION of METABOLIC ABNORMALITIES
- * **HYPOCALCEMIA**
 - ~ HYPOPARATHYROIDISM
 - ~ VITAMIN D DEFICIENCY
 - ~ $\uparrow\uparrow$ CALCIUM LEAVING the BLOOD
- * ELECTROLYTE IMBALANCES
- * DISORDERS AFFECTING ACID-BASE BALANCE

SYMPTOMS

- * NUMBNESS AROUND MOUTH
- * PAINFUL MUSCLE CRAMPS
- * PARESTHESIAS
- * LARYNGOSPASM
 - ~ LARYNGEAL STRIDOR
- * VOMITING
- * SEIZURES
- * $\downarrow\downarrow$ CARDIAC FUNCTION



Hypercalcemia signs & symptoms

www.medinaz.com



SYMPTOMS OF HYPERCALCEMIA

- Fatigue or weakness
- Feeling thirsty and frequent urination
- Constipation
- Nausea and vomiting
- Abdominal pain
- Confusion, irritability, or depression
- Bone pain or muscle weakness
- Irregular heartbeat or palpitations



Hypercalcemia

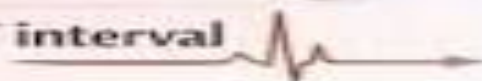
VS

Hypocalcemia



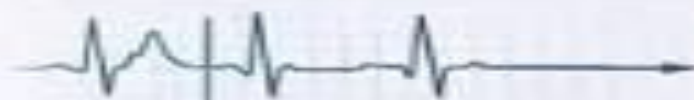
Hypercalcemia

- 'Stones, bones, groans, psychiatric overtones'
- Kidney stones
- Bone pain
- Constipation + abdominal pain
- Confusion / lethargy
- Shortened QT interval



Hypocalcemia

- Tetany (muscle spasms)
- Perioral numbness
- Seizures
- Chvostek & Trousseau signs
- Prolonged QT interval
- Irritability



Quick Clue:

High Ca = sluggish + stones

Low Ca = spasms + seizures

PHOSPHORUS

HYPOPHOSPHATEMIA

Definition

- Abnormally low levels of phosphate in the blood (< 2.5 mg/dL)

Causes

- Chronic alcoholism
- Malnutrition or starvation
- Refeeding syndrome
- Diabetic ketoacidosis (during treatment)
- Hyperparathyroidism
- Chronic antacid use (aluminum/magnesium-based)
- Respiratory alkalosis

Complications

- ▲ Impaired oxygen delivery (ATP production)
- ▲ Heart failure
- ▲ Hemolysis

Symptoms

- Muscle weakness
- Bone pain or fractures
- Confusion, irritability
- Paresthesia
- Difficulty breathing (respiratory muscle weakness)
- Seizures or coma (severe cases)



Treatment

- Oral phosphate supplements (mild cases)
- IV phosphate (severe or symptomatic cases)

Symptoms of **PHOSPHORUS** Deficiency 🧩



Weak Bones
/ Fractures



Muscle
Weakness



Fatigue



Confusion
or Anxiety



Poor Tooth
Formation



Loss of
Appetite



Numbness
or Tingling



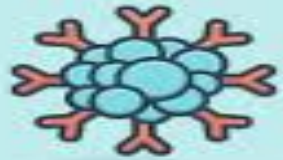
Anemia

Hyperphosphatemia

Causes of Hyperphosphatemia



Excessive dietary intake of phosphorus



Tumor lysis syndrome



Renal failure



Hypoparathyroidism
($\rightarrow$ hypocalcemia $\rightarrow$ hyperphosphatemia)



Signs & Symptoms

- **Most patients with hyperphosphatemia are asymptomatic they occasionally report hypocalcemic symptoms.**
- **Muscle cramps**
- **Tetany**
- **Joint pain**
- **Renal failure**
- **Anuria**
- **Hypertension**
- **Edema**
- **Cardiac arrest**

SYSTEMIC EFFECTS OF SEVERE HYPOPHOSPHATEMIA



Neuromuscular

- Weakness
- Paresthesia



Cardiovascular

- Decreased myocardial contractility
- Arrhythmias



Hematologic

- Impaired oxygen delivery
- Hemolysis



Neurological

- Confusion
- Seizures



Phosphorus Imbalance

SIGNS & SYMPTOMS



HYPERPHOSPHATEMIA



"Mice Always Dig Silently"

- M** Muscle spasms & tetany
- A** Arrhythmias
- D** Dry nails/skin
- S** Seizures

Directly linked
to **LOW** levels
of calcium
(hypocalcemia)



HYPOPHOSPHATEMIA



"A Lion Often Wanders"

- A** Arrhythmias
- L** Loss of appetite
- O** Osteoporosis
- W** Weakness, fatigue



Calcium supplements

